INCH-POUND
MIL-DTL-16377/73A
30 June 2014
SUPERSEDING
MIL-DTL-16377/73(SH)
27 September 2004

DETAIL SPECIFICATION SHEET

FIXTURES, LIGHTING; FLUORESCENT AND SSL EQUIVALENT, EMERGENCY LIGHTING, 20 WATTS, 120 VOLTS, 60 HERTZ, WATERTIGHT, SYMBOLS 331.1EM, 331.2EM, 347.2EM, 347.3EM, 331.1EMSSL, 331.2EMSSL, 347.2EMSSL, AND 347.3EMSSL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-16377.

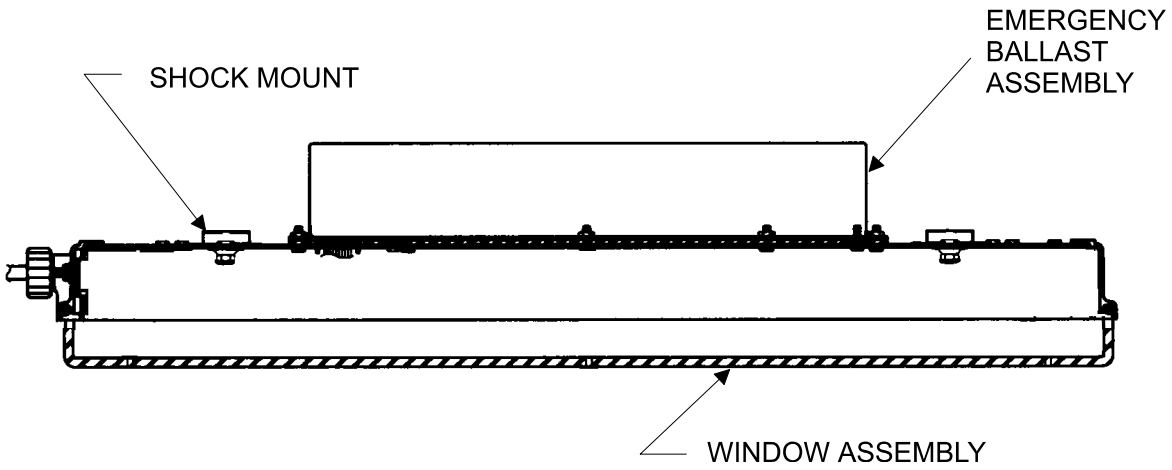
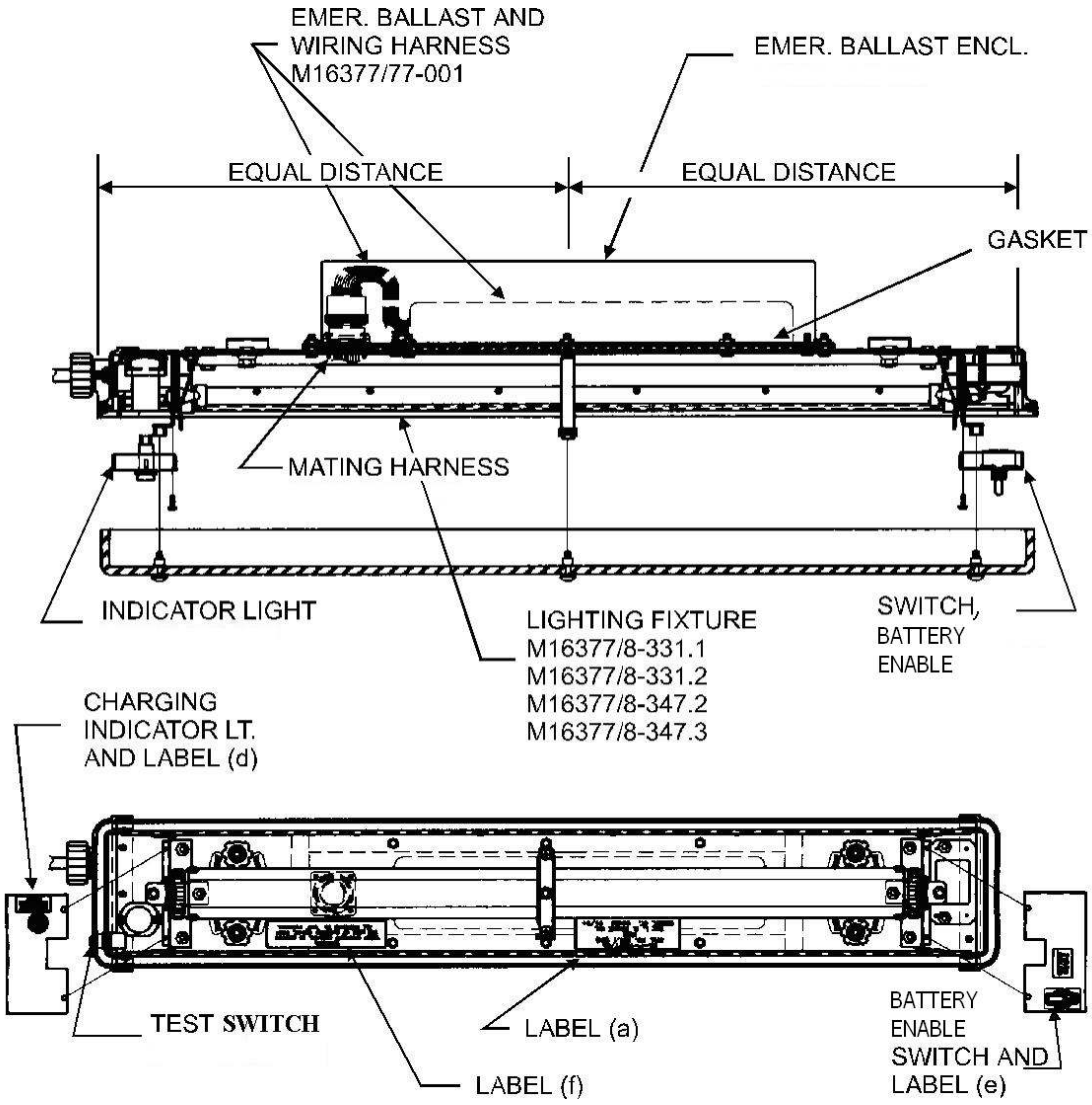


FIGURE 1. Typical outline of symbols 331.1EM, 331.2EM, 347.2EM, and 347.3EM.

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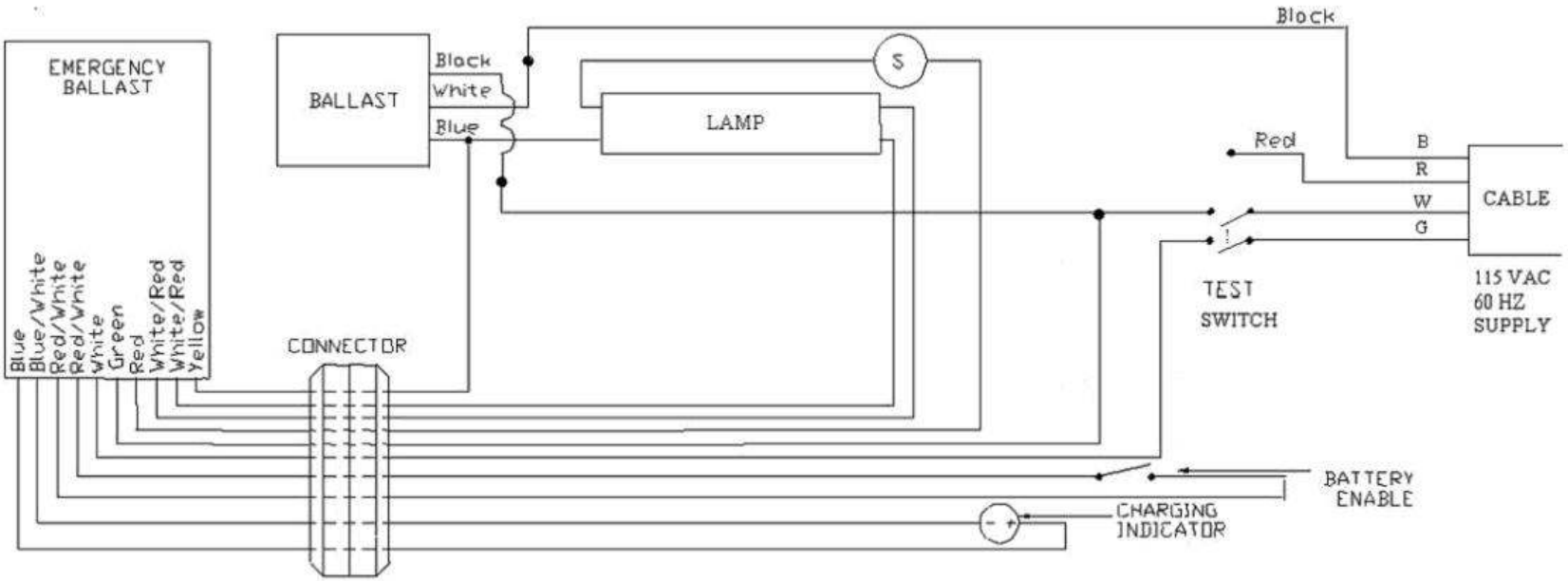


NOTES:

1. For dimensional data, see figures 1, 2, and 3 of MIL-DTL-16377/8.
2. For tolerances other than specified herein, see 3.5.2 of MIL-DTL-16377.

FIGURE 2. Typical view of symbols 331.1EM, 331.2EM, 347.2EM, and 347.3EM.

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NOTES:

1. Ballast black and white wiring not color coded with input cable black and white, emergency lamp only.
2. Emergency ballast white and green wiring not color coded with input cable white and green.

FIGURE 3. Wiring diagram.

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FLUORESCENT REQUIREMENTS:

CLASSIFICATION:

Type: I
Class: 2

SYMBOL NUMBERS:

331.1EM: Complete lighting fixture assembly, standard mounting, with one 20-watt cool white fluorescent lamp, white diffusing window assembly, and fluorescent emergency ballast.

331.2EM: Complete lighting fixture assembly, flush mounting, with one 20-watt cool white fluorescent lamp, white diffusing window assembly, and fluorescent emergency ballast.

347.2EM: Complete lighting fixture assembly, standard mounting, with one 20-watt cool white fluorescent lamp, clear and white prismatic window assembly, and fluorescent emergency ballast.

347.3EM: Complete lighting fixture assembly, flush mounting, with one 20-watt cool white fluorescent lamp, clear and white prismatic window assembly, and fluorescent emergency ballast.

DESIGN:

Electrical rating: 20 watts, 120 volts, 60 hertz.

Dimension and configurations: See figures 1, 2, and 3 of MIL-DTL-16377/8 and figure 10 of MIL-DTL-16377.

Photometric data: See figure 4 of MIL-DTL-16377/8.

Lampholder spacing: See figure 21 of MIL-DTL-16377.

Wiring diagram: See figure 3.

Mounting dimensions:

Symbols 331.1EM and 347.2EM: See figure 25 of MIL-DTL-16377.

Symbols 331.2EM and 347.3EM: See figure 26 of MIL-DTL-16377.

Enclosure: Watertight. See 3.5.11 of MIL-DTL-16377.

Cable: 10 feet of Type LSFHOF-3 in accordance with MIL-DTL-24643/3.

Gasket: Neoprene rubber (durometer 35-45).

Weight:

Symbols 331.1EM and 347.2EM: 13.5 pounds maximum.

Symbols 331.2EM and 347.3EM: 15 pounds maximum.

Designation and marking: The following labels shall be furnished in accordance with MIL-DTL-16377. Labels (a), (b), and (c) in accordance with MIL-DTL-16377/8. Labels (d), (e), and (f) as shown on figure 2:

a. Two Type I labels.

b. One Type II label for Symbols 331.2EM and 347.3EM only.

c. One Type IV label. The information to be shown on the Type IV label shall be “USE THE FOLLOWING REPAIR PARTS ONLY:” followed by a list of all standard parts shown in table I by name and stock number only.

d. One Type IV label. The information to be shown on the Type IV label shall be “CHARGING INDICATOR LIGHT.”

e. One Type IV label. The information to be shown on the Type IV label shall be “BATTERY ENABLE.”

f. One Type V label. The information to be shown on the Type V label shall be “DURING MAINTENANCE, PROTECTIVE DEVICE MUST BE OPENED OR PULLED TO SECURE AC POWER TO THIS FIXTURE AND THE BATTERY ENABLE SWITCH MUST BE IN THE OFF POSITION.”

Standard parts: The standard parts and their respective part numbers shown in table I shall be used in the construction of the fixtures specified herein.

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SSL REQUIREMENTS:

CLASSIFICATION:

Type: III
Class: 2

SYMBOL NUMBERS:

331.1EMSSL:	Complete lighting fixture assembly, standard mounting, with LED light source, white diffusing window assembly, and emergency battery back-up.
331.2EMSSL:	Complete lighting fixture assembly, flush mounting, with LED light source, white diffusing window assembly, and emergency battery back-up.
347.2EMSSL:	Complete lighting fixture assembly, standard mounting, with LED light source, clear and white prismatic window assembly, and emergency battery back-up.
347.3EMSSL:	Complete lighting fixture assembly, flush mounting, with LED light source, clear and white prismatic window assembly, and emergency battery back-up.

DESIGN:

Electrical rating:	20 watts maximum, 120 volts, 60 hertz.
Dimension and configurations:	External dimensions shall not exceed those shown on figures 1, 2, and 3 of MIL-DTL-16377/8 and figure 10 of MIL-DTL-16377.
Photometric data:	See figure 4 and table I of MIL-DTL-16377/8 and 3.7.1 of MIL-DTL-16377.
Emergency mode lumen output:	Luminaire shall provide a minimum of 240 lumens on a fully charged battery and maintain over 144 lumens over a 5-hour runtime.
Chromaticity:	See 3.7.4 of MIL-DTL-16377.
Fail-safe circuit:	See 3.17 of MIL-DTL-16377.
Wiring:	Shall maintain same control scheme as legacy fluorescent, including battery enable switch, battery charging indicator.
Mounting dimensions:	
Symbols 331.1EMSSL and 347.2EMSSL:	See figure 25 of MIL-DTL-16377.
Symbols 331.2EMSSL and 347.3EMSSL:	See figure 26 of MIL-DTL-16377.
Enclosure:	Watertight. See 3.5.11 of MIL-DTL-16377.
Cable:	10 feet of Type LSFHOF-3 in accordance with MIL-DTL-24643/3.
Gasket:	Neoprene rubber (durometer 35-45), if design uses legacy fluorescent emergency ballast enclosure.
Weight:	
Symbols 331.1EMSSL and 347.2EMSSL:	13.5 pounds maximum.
Symbols 331.2EMSSL and 347.3EMSSL:	15 pounds maximum.
Designation and marking:	The following labels shall be furnished in accordance with MIL-DTL-16377. Labels (a), (b), and (c) in accordance with MIL-DTL-16377/8. Labels (d), (e), and (f) as shown on figure 2:

a. Two Type I labels.
b. One Type II label for Symbols 331.2EM and 347.3EM only.

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c. One Type IV label. The information to be shown on the Type IV label shall be "USE THE FOLLOWING REPAIR PARTS ONLY:" followed by:
   (1) A list of all standard parts shown in table I by name and stock number only, if appropriate, or
   (2) A list of available replacement parts and stock numbers, or if unavailable, manufacturer's part numbers.

d. One Type IV label. The information to be shown on the Type IV label shall be "CHARGING INDICATOR LIGHT."

e. One Type IV label. The information to be shown on the Type IV label shall be "BATTERY ENABLE."

f. One Type V label. The information to be shown on the Type V label shall be "DURING MAINTENANCE, PROTECTIVE DEVICE MUST BE OPENED OR PULLED TO SECURE AC POWER TO THIS FIXTURE AND THE BATTERY ENABLE SWITCH MUST BE IN THE OFF POSITION."

BATTERY BACK-UP SYSTEM REQUIREMENTS:

Battery back-up system for Type III luminaires.

DESIGN:

Assembly:	The battery back-up system consists of a high temperature, rechargeable battery; battery charger; AC power sensing circuit; wiring harness; enclosure.
Circuit design:	Back-up system shall incorporate Thyristor devices for protection against high voltage spikes.
Battery back-up location:	Battery back-up assembly shall fit within the dimensions of figure 1 of MIL-DTL-16377/8 or within the legacy fluorescent emergency ballast enclosure, as described in table I.
Battery material:	Use of other than NI-CAD battery systems requires specific authorization from NAVSEA.
Emergency operation time:	5 hours minimum.
Full recharge time:	20 hours maximum.
Temperature rating:	131 °F (55 °C) minimum.
Charging power:	4.0 watts maximum.
System power factor:	>0.95.
System harmonics:	<3 percent.
Wiring:	Connection between battery back-up system and the rest of the luminaire shall be made with a commercial wiring harness that allows for easy replacement of battery back-up system.

EXAMINATION AND TESTS:

Examination and tests shall be as specified in MIL-DTL-16377 and as follows:

First article inspection:	
Complete fixtures:	See table II for Type I fixtures and table III for Type III luminaires.
Conformance:	
Comparison inspection:	Same as first article inspection.
Conformance inspection:	See MIL-DTL-16377.

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TABLE I. Standard parts for type I fixtures.

Name	Military P/N or Specification	National Stock Number (NSN)	Remarks
Complete fixture, Symbol 331.1	M16377/8-331.1	6210-00-548-0598	For use with Symbol 331.1EM
Complete fixture, Symbol 331.2	M16377/8-331.2	6210-01-088-2511	For use with Symbol 331.2EM
Complete fixture, Symbol 347.2	M16377/8-347.2	6210-01-088-6945	For use with Symbol 347.2EM
Complete fixture, Symbol 347.3	M16377/8-347.3	6210-01-088-6186	For use with Symbol 347.3EM
Window, white diffusing	M16377/8-001	6210-00-548-0601	
Window, clear and white prismatic	M16377/8-003	6210-01-438-2960	
Mating harness	AMP part No. 206043-1 or equal		Mates with M16377/77-001
Emergency ballast and wiring harness	M16377/77-001	6250-01-491-7039	
Indicator light	Commercial	6220-01-415-7889	LED type
Battery enable switch[1/]	Commercial	5930-01-414-9240	DPST
Enclosure for emergency ballast			Aluminum; approximately 3″ H x 3 ½″ W x 16″ L
Test switch	Commercial	5930-01-416-0342	Double pole, test switch with rubber boot
Switch cover, rubber boot	MIL-DTL-5423/2		Neoprene
Red filter tube	M16377/42-003	6210-00-295-2838	Not furnished, see 3.7.9 of MIL-DTL-16377
Broad band blue filter tube	M16377/42-005	6210-00-441-5138	Not furnished, see 3.7.9 of MIL-DTL-16377
Amber filter tube	M16377/42-006	6210-01-166-7184	Not furnished, see 3.7.9 of MIL-DTL-16377
Gray filter tube (1% transmission)	M16377/42-007	6210-01-498-8001	Not furnished, see 3.7.9 of MIL-DTL-16377
Gray filter tube (2.5% transmission)	M16377/42-009	6210-01-313-8168	Not furnished, see 3.7.9 of MIL-DTL-16377

NOTE:

1/ Switch shall normally be left in the “Battery Enable” position.

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TABLE II. First article and comparison inspection for complete type I fixtures. [1/]

Test	Requirement	Test Method	Remarks
Examination	3.13	4.5	
Operation	3.13	4.8.1	
Power interface	3.6.18	4.8.21	
Power quality	3.6.18.1	4.8.22	
Voltage transient spike	3.6.18.2	4.8.23	
Electromagnetic interference	3.6.14	4.8.16	
Dielectric withstanding voltage	3.6.15	4.8.2	[2/]
Insulation resistance	3.6.16	4.8.3	[2/]
Enclosure effectiveness	3.5.11	4.8.14.4	[2/, 3/]
Shock	3.5.12	4.8.7	[2/]
Vibration	3.5.13	4.8.8	[2/]
Enclosure effectiveness	3.5.11	4.8.14.4	[2/, 3/]
Dielectric withstanding voltage	3.6.15	4.8.2	[2/]
Insulation resistance	3.6.16	4.8.3	[2/]
Noise	3.5.14	4.8.9	[2/]
Continuity of grounding	3.6.13.4	4.8.12	
Light output	3.7.1	4.8.6.1	Candlepower
Optical uniformity	3.7.7.1	4.8.4	
Stress relief	3.8.5.3	4.8.5	
Magnetic permeability	3.4.8	4.8.18	
Emergency light output	3.7.1.1	4.8.6.8	Lumen output
Emergency ballast performance: Operating time and discharge test	3.6.19.3	4.8.26	

NOTES:

[1/] Requirement and test method paragraph numbers refer to MIL-DTL-16377.

[2/] In accordance with 4.3.1 of MIL-DTL-16377, tests shall be performed on one luminaire in the order specified.

[3/] Watertight.

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TABLE III. First article and comparison inspection for complete type III luminaires. 1/

Test	Requirement	Test Method	Remarks
Examination	3.13	4.5	
Operation	3.13	4.8.1	
Power interface	3.6.18	4.8.21	
Power quality	3.6.18.1	4.8.22	
Voltage transient spike	3.6.18.2	4.8.23	
Electromagnetic interference	3.6.14	4.8.16	
Dielectric withstanding voltage	3.6.15	4.8.2	2/
Insulation resistance	3.6.16	4.8.3	2/
Enclosure effectiveness	3.5.11	4.8.14	2/, 3/
Shock	3.5.12	4.8.7	2/
Vibration	3.5.13	4.8.8	2/
Enclosure effectiveness	3.5.11	4.8.14	2/, 3/
Dielectric withstanding voltage	3.6.15	4.8.2	2/
Insulation resistance	3.6.14	4.8.3	2/
Noise	3.5.14	4.8.9	2/
Continuity of grounding	3.6.13.4	4.8.12	
Photometrics	3.7.1	4.8.6	Candlepower
Brightness	3.7.5	4.8.6.4	
Chromaticity	3.7.4	4.8.6.5	
Correlated color temperature (CCT)	3.7.4.1.1	4.8.6.6	
Lumen maintenance	3.7.11	4.8.20	
Fail-safe circuit design	3.17	4.8.24	
Optical uniformity	3.7.7.1	4.8.4	
Stress relief	3.8.5.3	4.8.5	
Magnetic permeability	3.4.8	4.8.18	
Emergency light output	3.7.1.1	4.8.6.8	Lumen output
Emergency battery back-up performance: Operating time and discharge test	3.6.19.3	4.8.26	

NOTES:

1/ Requirement and test method paragraph numbers refer to MIL-DTL-16377.

2/ In accordance with 4.3.1 of MIL-DTL-16377, tests shall be performed on one luminaire in the order specified.

3/ Watertight.

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TABLE IV. Part number and general information.

Military Part Number M16377/73-	Item Name	National Stock Number (NSN)
331.1EM	Complete fixture, Symbol 331.1EM	6210-01-477-4611
331.2EM	Complete fixture, Symbol 331.2EM	6210-01-508-6405
347.2EM	Complete fixture, Symbol 347.2EM	
347.3EM	Complete fixture, Symbol 347.3EM	
331.1EMSSL	Complete fixture, solid state, Symbol 331.1EMSSL	
331.2EMSSL	Complete fixture, solid state, Symbol 331.2EMSSL	
347.2EMSSL	Complete fixture, solid state, Symbol 347.2EMSSL	
347.3EMSSL	Complete fixture, solid state, Symbol 347.3EMSSL	

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
Navy – SH
DLA – IS

Preparing activity:
Navy – SH
(Project 6210-2012-033)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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